② / 2: $\frac{73}{97} \bmod 357$

$$\left(\begin{array}{c|c} 357 & 0 \\ 97 & 1 \\ 0 & 0 \\ \leq 73 & \end{array}\right) \overset{M_{12}(-1)}{\sim} \left(\begin{array}{c|c} 260 & -1 \\ 97 & 1 \\ 0 & 0 \\ \leq 73 & \end{array}\right) \overset{M_1(\frac{1}{2})}{\sim} \left(\begin{array}{c|c} 130 & 178 \\ 97 & 1 \\ 0 & 0 \\ \leq 73 & \end{array}\right) \overset{M_1(\frac{1}{2})}{\sim}$$

$$\sim \left(\begin{array}{c|c} 65 & 89 \\ 97 & 1 \\ 65 & 89 \\ \leq 8 & \end{array}\right) \overset{M_{12}(-1)}{\sim} \left(\begin{array}{c|c} 65 & 89 \\ 32 & -88 \\ 65 & 89 \\ \leq 8 & \end{array}\right) \overset{M_2(\frac{1}{2})}{\sim} \left(\begin{array}{c|c} 65 & 89 \\ 16 & -44 \\ 65 & 89 \\ \leq 8 & \end{array}\right) \overset{M_2(\frac{1}{2})}{\sim} \left(\begin{array}{c|c} 65 & 89 \\ 8 & -22 \\ 65 & 89 \\ \leq 8 & \end{array}\right)$$

$$\overset{M_{23}(1)}{\sim} \left(\begin{array}{c|c} 65 & 89 \\ 8 & -22 \\ 73 & 67 \\ \leq 0 & \end{array}\right)$$

Ombeem: 67

Проверка:

$67 \cdot 97 \bmod 357 = 73$

Задача 1: $(\lambda^3 + 4\lambda + 2)^{-1} \bmod (\lambda^4 + 2\lambda^2 + 3, 7)$

$(1,0,4,2)$ $\qquad$ $(1,0,2,0,3)$

$$\left(\begin{array}{c|c} (0,1,0,4,2) & 1 \\ (1,0,2,0,3) & 0 \end{array}\right) \overset{M_2(3)}{\sim} \left(\begin{array}{c|c} (0,1,0,4,2) & 1 \\ (3,0,6,0,9) & 0 \end{array}\right) \overset{M_{21}(-1)}{\sim}$$

$$\left(\begin{array}{c|c} (0,1,0,4,2) & 1 \\ (3,-1,6,-4,7) & -1 \end{array}\right) \equiv \left(\begin{array}{c|c} (0,1,0,4,2) & 1 \\ (3,6,6,3,0) & 6 \end{array}\right) \equiv \left(\begin{array}{c|c} (0,1,0,4,2) & 1 \\ (3,6,6,3,0) & 5,0,3,0,0 \end{array}\right)$$

$$M_2\left(\tfrac{1}{2}\right)\left(\begin{array}{cc|c} (1,0,4,2) & & 1 \\ (3,6,6,3) & & (5,0,3,0) \end{array}\right) M_2(3)\left(\begin{array}{cc|c} (1,0,4,2) & & 1 \\ (9,18,18,9) & & (15,0,9,0) \end{array}\right) \sim$$

$$M_{12}(-1)\left(\begin{array}{cc|c} (1,0,4,2) & & 1 \\ (1,4,0,0) & & (1,0,2,6) \end{array}\right) \equiv \left(\begin{array}{cc|c} (1,0,4,2) & & 1 \\ (1,4,0,0) & & (5,1,3,2,0) \end{array}\right)$$

$$M_2\left(\tfrac{1}{2}\right)\left(\begin{array}{cc|c} (1,0,4,2) & & 1 \\ (0,1,4,0) & & (5,1,3,2) \end{array}\right) \equiv \left(\begin{array}{cc|c} 1,0,4,2 & & 1 \\ 0,1,4,0 & & 4,5,2,3,0 \end{array}\right) \sim$$

$$M_2\left(\tfrac{1}{2}\right)\left(\begin{array}{cc|c} (1,0,4,2) & & 1 \\ (0,0,1,4) & & (4,5,2,3) \end{array}\right) M_1(2)\left(\begin{array}{cc|c} 2,0,8,4 & & 2 \\ 0,0,1,4 & & 4,5,2,3 \end{array}\right) M_{12}(-1)$$

$$\sim \left(\begin{array}{cc|c} 2,0,0,0 & & 2 \\ 0,0,1,4 & & 4,5,2,3 \end{array}\right) M_{12}(-1)\left(\begin{array}{cc|c} 2,0,0,0 & & 3,2,5,6 \\ 0,0,1,4 & & 4,5,2,3 \end{array}\right) \equiv$$

$$\equiv \left(\begin{array}{cc|c} 2,0,0,0 & & 3,2,5,6 \\ 0,0,1,4 & & 4,5,2,3 \end{array}\right) \equiv \left(\begin{array}{cc|c} 2,0,0,0 & & 5,3,5,5,0 \\ 0,0,1,4 & & 0,4,5,2,3 \end{array}\right) M_1\left(\tfrac{1}{2}\right)$$

$$\left(\begin{array}{ccc|cccc} 2,0,0 & 5,3,5,5 \\ 0,1,4 & 4,5,2,3 \end{array}\right) \equiv \left(\begin{array}{ccc|cccc} 2,0,0 & 3,5,2,5,0 \\ 0,1,4 & 4,5,2,3 \end{array}\right) M_1\left(\tfrac{1}{2}\right) \left(\begin{array}{cc|ccc} 2,0 & 3,5,2,5 \\ 1,4 & 4,5,2,3 \end{array}\right)$$

$$\equiv \left(\begin{array}{cc|ccccc} 2,0 & 5,3,4,2,0 \\ 1,4 & 0,4,5,2,3 \end{array}\right) M_1\left(\tfrac{1}{2}\right) \left(\begin{array}{cc|cccc} 0,2 & 3,3,4,2 \\ 4,2 & 2,6,1,5 \end{array}\right) M_1(4) \left(\begin{array}{cc|ccc} 0,1 & 5,5,2,1 \\ 4,2 & 2,6,1,5 \end{array}\right)$$

$$2^{(4,0,2,0,3),7} \equiv (3,3,4,2) \mid \cdot 2^{-1} \bmod 7 ?$$

Проверка:

$$(0,1,0,4,2)\cdot(0,5,5,2,1) \bmod (1,0,2,0,3)$$

	5,5,2,1	0	6,6,1,4	3,3,4,2
1,0	5,5,2,1	5,5,2,1,0 −5,0,3,0,1 5,6,1,6	5,6,1,6,0 −5,0,3,0,1 0,5,6,6 +0,6,1,4 5,4,0,3	5,4,0,3,0 −5,0,3,0,1 4,4,3,6 +3,3,4,2 0,0,0,1

Ответ: 1